

Seeking Trustworthy Information on COVID-19: A Qualitative Study with the African American Community

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ABSTRACT

COVID-19 has been known to have a disproportionate impact on African Americans in the United States. Although studies have been conducted on the reasons for this disparity, there has been less focus on how the African American population sought trustworthy health information using technology. This is important because African Americans' mistrust of the medical system has been suggested as a possible reason for the disproportionate impact. Therefore, we conducted interviews with 18 African Americans with chronic conditions to discover what types of challenges they faced while searching for trustworthy information on COVID-19 and the strategies they used to overcome these challenges. We found that participants actively evaluated the credibility of different information sources, searched for first-hand information from people they could relate to, and tried to avoid or reduce media consumption to prevent information overload.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods; • **Social and professional topics** → *Race and ethnicity*.

KEYWORDS

Black American community, chronic condition, trust, COVID-19, in-depth interview

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1 INTRODUCTION

Although COVID-19 has had a global impact, it is especially known to disproportionately impact African Americans in the United States [7, 34]. In the U.S., minority-majority counties that consist mostly of African Americans have three times the rate of infections and six times the fatalities than counties with Caucasian majority populations [21]. Several reasons have been suggested as to why African Americans are more vulnerable to COVID-19. One reason is that they tend to have higher than average rates of chronic health conditions [1, 4, 6]. Chronic health conditions such as diabetes [4], obesity [18], hypertension [6], and chronic obstructive pulmonary disease [1] are more likely to result in more severe COVID-19 cases. In addition, structural racism generates inequity in social risk factors (e.g. homelessness [27], and higher poverty rates that lead to limited healthcare access [15, 28], which in turn results in higher rates of COVID-19 infections. The medical community has also suggested a high level of medical mistrust of the U.S. healthcare system and the government as a reason for higher rates of infection. Medical mistrust has resulted in the population's hesitancy in receiving a COVID-19 vaccine [2] and prevented them from seeking medical care or prevention guidelines regarding COVID-19 [2].

However, there has been a lack of research on how African Americans search for and use trustworthy information on COVID-19. African Americans rely heavily on the use of smartphones and the internet to seek health information [12]. Moreover, African American Twitter communities have shared COVID-19 prevention behaviors, indicating their usage of social media to seek and disseminate health information [31]. In searching for health information online, it is essential to understand how trustworthy information is searched for and processed, especially regarding COVID-19. Because COVID-19 was new and unfamiliar to everyone, it was crucial to convey and obtain trustworthy information. This was especially true for African Americans with chronic conditions because they were at higher risk than others and exhibited their inherent medical mistrust [26, 36].

We extend the literature on the health information seeking behavior of African Americans in the following ways. First, we examine what challenges the African American population with chronic conditions faced while seeking trustworthy COVID-19 information online. Second, we investigate the strategies they used to search for and process trustworthy information using technology. Exploring their challenges and strategies is important because by understanding their behaviors for seeking trustworthy information, we can

Table 1: Demographic information of the participants

Participant	Gender	Age	Chronic Condition	Occupation	Household Income
P01	Female	45	Diabetes	Transit coach operator	\$20,000-\$34,999
P02	Female	44	Multiple sclerosis, pre-diabetes, depression	Unemployed	\$35,000-\$49,999
P03	Female	43	Multiple sclerosis, high blood pressure	Retired	\$50,000-\$74,999
P04	Female	55	Asthma, hypertension, high cholesterol	Unemployed	below \$20,000
P05	Male	54	Degenerative bone disease, high blood pressure	Unemployed	below \$20,000
P06	Female	55	Diabetes, asthma, COPD	Computer repair	below \$20,000
P07	Female	62	Hypertension, bipolar, degenerative disc	Receptionist	\$20,000-\$34,999
P08	Female	24	Asthma, chronic back pain	Homemaker	below \$20,000
P09	Male	58	Hypertension	Software engineer	over \$75,000
P10	Female	22	Asthma	Student	Below\$20,000
P11	Female	62	Rheumatoid arthritis	Referral coordinator	\$35,000-\$49,999
P12	Female	42	Diabetes, tricuspid atresia	Unemployed	over \$75,000
P13	Female	21	Scoliosis, asthma, hypertension	Student	below \$20,000
P14	Female	53	Obesity, high blood pressure	HR manager	over \$75,000
P15	Female	50	Asthma	Direct care worker	below \$20,000
P16	Female	63	COPD	Unemployed	below \$20,000
P17	Female	50	Autoimmune condition, arthritis	Unemployed	below \$20,000
P18	Female	49	Hypertension, high cholesterol	Caseworker	\$35,000-\$49,999

suggest new technology design implications which can expand beyond the pandemic to shape their attitudes and behaviors regarding vaccinations and other medical treatments. As the subject of our study, we especially focus on African American individuals with chronic conditions, as they are the population at highest risk from COVID-19.

2 RELATED WORKS

In the healthcare domain, mobile health technology (mHealth) has been useful in providing health information for the African American population [5, 17] because it can address the challenges they face such as difficulties in attending office appointments due to mobility issues or time limitations [13]. MHealth, the recently developed delivery of healthcare intervention through smartphones, has heavily focused on providing personalized interventions [22]. For instance, researchers have designed applications that could send tailored messages to African Americans with high blood pressure to increase medication adherence [3] or track blood glucose levels and receive feedback [19]. These technologies have been shown to be effective in promoting healthy behaviors, recording patients' blood pressure levels, and inducing better adherence to medications.

Recent studies have also examined the importance of understanding the socioeconomic and cultural backgrounds of the African American population in developing health technologies. Several of these studies incorporated participatory design methodologies to understand their challenges as well as needs [23] and design technology that can address their needs [16, 37]. Others focused on understanding how underserved communities face difficulties in obtaining healthcare and health information [8] or the importance of trust in developing an intervention to prevent human immunodeficiency virus (HIV) based on focus group interviews with African American youth [32].

Trust has been proven to be an important factor in determining how often users interact with systems providing healthcare services [35], which influences their acceptance of health information [25], adherence to medical advice and medication [20], and satisfaction with the system [9]. While trust towards the technology has mainly been assessed using surveys [24, 29], interviews have also been used to evaluate how and why users trust the system [11]. Studies have identified the factors that lead to trust towards health information technology such as understandability, relevance, usefulness, ease of use, and interactivity [38]. Higher levels of these factors have contributed to higher levels of trust, which led to a stronger intention to use the health intervention technology in making health decisions [38].

3 METHODOLOGY

We remotely interviewed 18 African Americans with chronic conditions in the U.S. The study's participants included: 16 women and 2 men with a mean age of 47.3. The interviews were conducted between November 2020 and March 2021, and participants were recruited through a university's health clinic and its recruitment system. We sent out emails and text messages to search for those who wanted to participate. To participate in the study individuals had to (1) be African American with one or more chronic conditions that pose a high risk to COVID-19 patients, (2) own a laptop or a desktop with a webcam and have an internet connection at home to participate in the Zoom study, and (3) be between 18 and 65 years old. Once these eligibility criteria were confirmed, the first author sent out emails to plan and conduct Zoom interviews. Table 1 summarizes the demographic information of our participants. This study was approved by the university's Institutional Review Board for Medical Research.

The semi-structured interviews were conducted by the first and the second author and lasted 60 to 70 minutes. The questions focused on participants' challenges during COVID-19 and their trust towards their clinicians or institutions such as the CDC or other governmental healthcare organizations. Participants were asked specific questions regarding the type(s) of chronic conditions they have, their challenges of managing them during the pandemic, opinions on how the medical system and the government have handled the pandemic, approaches to searching for trustworthy COVID-19 information, and their attitudes towards the COVID-19 vaccine (e.g., sample questions included, "What is the most challenging or stressful thing you experience in managing your health condition during this pandemic?", "How did you try to look for trustworthy information on COVID-19?"). Participants received \$20 dollars as incentives for taking part within the study. All interviews were recorded upon receiving consent from the participants. We later transcribed the recordings through a third-party service that was approved by our institution for HIPAA.

For the analysis, two researchers open-coded the first five interviews to identify central themes using affinity diagrams [30]. We started by broadly exploring common themes and gradually focused on the themes of the challenges participants faced during COVID-19 in obtaining trustworthy information and the technologies and strategies they used to search for credible information. This helped us restructure our interview questions and concentrate on trust in subsequent interviews. After this process, four members of the research team met weekly to develop identified themes on the reasons for trust. In a grounded theory approach, axial coding was conducted to identify the relationships between themes [30]. The themes produced by the researchers were compared, discussed, and revised through a series of discussions until agreements were reached. In this paper, we report our findings on the three challenges faced by the African Americans with chronic conditions in gathering trustworthy COVID-19 information and the strategies they used to overcome those challenges.

4 FINDINGS

4.1 Actively evaluating the credibility of a variety of information sources

The first challenge of searching for trustworthy information on COVID-19 was being misled by false or partial information. We found that participants were either deceived by false information on the internet or perceived bias in information from certain online sources. For instance, one of our participants had a false belief that the ventilators made her spouse die from COVID-19, which led her to search Google for more information about the ventilators.

"He got COVID when he was at the rehab center and what they were doing was just managing with oxygen and then when he got rushed to the hospital, then they decide within a day 'we're going to put him on a ventilator' and I just believe that's what killed him. So that's what made me intrigued to look up, Is that necessary, do everyone have to be on a ventilator for COVID-19. And it came up that that wasn't necessarily the case" (P3, female).

When P3 performed a search using Google, she went to random websites or blogs that appeared instead of websites perceived by many as reputable and trustworthy (e.g. the CDC) [33] because she was unfamiliar with such sources. This led her to encounter information that downplayed the necessity of ventilators for people at higher risk. This information strengthened her prior belief that the ventilators and the hospitals were responsible for killing her husband.

Participants also perceived bias in information from certain sources, such as social media, because they reflect subjective views of individuals. For instance, participants categorized Facebook as an "untrustworthy information source" due to its bias and misinformation. Because Facebook was a platform based on people's personal opinions, participants felt that its information lacked objectivity.

"Certain stuff on Facebook is fake and how it's being said or something. Some news stuff not always (trustworthy) because to me the news isn't always believable, they just put what they want you to believe" (P8, female).

P8 believed that people on the social media site disseminate misinformation which signifies her general mistrust of people and the press. She believed that they could be deliberately propagating misinformation to confuse the public. Part of this comes from the general mistrust towards the public, believing that people spread such misinformation to take advantage of others.

Some participants also believed that the news media often delivered partial information, which made it harder for them to perceive it as a trustworthy source. For them, the news only focused on the negatives of COVID-19, such as the increasing number of COVID cases and its associated deaths, which caused people to become scared and depressed. Furthermore, it led participants to feel that there was a certain gap in information that the media did not address properly. They felt that important information was being overshadowed by the dramatized news of death and infection. They explained that the news was delivering simply what was shown or partial information of what was actually happening, without further discussion on specific why's and how's.

"So it's like if I'm watching or reading the news it's like I see what they're telling me but I know there's more behind it" (P8, female).

The media's emphasis on the number of cases and deaths left participants to feel a gap in information when they needed more information on where COVID-19 came from and how it affected the public.

In order to overcome this challenge, many of the participants actively explored a variety of information sources and evaluated their credibility to determine which specific sources they thought were trustworthy. Although participants kept their eyes on the mainstream news due to its accessibility, participants felt more trust towards government-related sources such as the CDC or medical sources (e.g., WebMD). The perceived credibility of these sources was a result of their long experience of using these sources as chronic condition patients. Their familiarity with these sources due to the sources' expertise on chronic conditions also made them trustworthy regarding COVID-19 information.

In addition, some participants looked for more objective and verified information through expert knowledge by reading scientific journals or statistics. They wanted information that was verified through scientific research or data that could be interpreted by themselves directly. They did not want to simply believe what the news was telling them, so they searched for and interpreted information on their own. For instance, P10 was a female college student who was more interested in interpreting data and charts on her own instead of mindlessly listening to what the news was telling her. Because she believed that not all information filtered through the media is credible, she chose to find raw data and extract meaning herself.

Although many participants relied heavily on medical-oriented sources, they wanted to evaluate and weigh the credibility of different sources and ensure that the information they were getting was factual and trustworthy. This was done through corroborating with different resources, which is demonstrated in the quote from P5. P5 had bone degenerative disease and high blood pressure, which made him an avid researcher of health-related information. His willingness to research became useful when evaluating the credibility of information on COVID-19.

"I first start off with whoever's given me the information and check if they're credible from the beginning. Then I can say that I trust a little bit more about them. If they've been uncredible and given off false information then I think more about that...I look up that name and see how credible the source is." (P5, male).

P5 performed an active and thorough research to identify dependable resources and confirm their credibility. Whenever he encountered a new source of information, he checked the legitimacy of the source by assessing its credibility. In this way, he constantly weighed and evaluated different sources by checking what kind of trustworthy or untrustworthy information they offer.

Our participants noted that they corroborated the information through other sources as well such as their primary clinicians, work colleagues, or family and friends. In this way, they could determine which sources were most trustworthy and rely on those sources for COVID-19 information.

4.2 Searching for first-hand experience by those who participants could relate to

Another challenge of participants is that they felt there was a gap in information because the information they obtained about COVID-19 or the vaccine was not closely tied to their health situation or their historical context. The first gap they addressed was how the government focused more on COVID-19 and put less emphasis on chronic conditions.

"Since this pandemic has hit, all the emphasis is on how many people have died as a result of this, as if people are no longer dying of anything else, as if there are no other ailments that people suffer from" (P9, male).

As can be seen from the quote above from P9, some participants believed that the previous government was disseminating propaganda, making it seem like COVID-19 is the only disease that mattered. They described that patients with chronic conditions were not being cared for because the government intentionally focused on COVID-19 and paid less attention to other chronic conditions.

Participants also felt there was a lack of information on how the COVID-19 vaccine was developed and distributed. This lack of information was especially problematic for our participants because many of them had misgivings about the vaccine due to their cultural and historical background. Specifically, many participants brought up issues of past discrimination and misinformation on the part of the government.

"So, the medical researchers have a history of annihilating people, simple as that. There was some other people rebutting (the vaccine) and saying, 'Don't do it. You know what happened with the Tuskegee study'" (P16, female).

As can be seen from the quote by P16, some participants mentioned the Tuskegee Syphilis Study when they discussed their hesitation in taking the vaccine. The Tuskegee study was conducted by the U.S. Public Health Service to study the effects of untreated syphilis among African American men [14]. It is now widely known to be unethical because the government purposely injected syphilis into participants but did not provide penicillin that would treat them, which ended up in participants' either dying from the disease or related complications. Participants described that this study's story has passed through generations among African Americans, making them more cautious about medical studies. This cautiousness and suspicion resulted in a wider gap in information because they wanted and needed more information to trust and take the vaccine.

The strategy that participants developed to obtain trustworthy information was to search for first-hand information, especially information from people of the same ethnicity. This was because those people shared the same cultural and historical background. Therefore, social media sources such as Facebook, TikTok, or Reddit were important to them because they provided information directly by people within their ethnicity, especially with regards to the COVID-19 vaccine. Participants felt more trust towards video contents in particular, which is a popular communication method in recent social media platforms such as TikTok [10]. This was because those videos were viewed as unedited and raw content recorded by smartphones. To our participants, the actual statements spoken by people on their social network sites or the videos by those who took the vaccine and explained their direct experience carried more value than news articles, especially if they were delivered by people with the same ethnicity.

"I have to mention he [the clinician] was African American. Now, I have a girlfriend who is Caucasian, and she took the vaccine. She didn't have no side effects. So, I said, 'Okay, well ...' The African American doctor went on to say that his side effects from the vaccine could no way at all compare to what COVID actually was, and he said, 'I implore you to get the vaccine.' His presentation was very compelling, and

that's what made me decide to go ahead and get it"
(P16, female).

As the quote from P16 suggests, even within social media, participants were influenced by clinicians of the same ethnicity with regards to the perception of the COVID-19 vaccine's safety. Whereas they did not trust Caucasian clinicians and medical professionals when they addressed the safety of the vaccine, they trusted it when it was delivered by African American clinicians. Being in the same ethnicity group allowed participants to feel that clinicians could relate to the lack of trust felt by the African American community towards the vaccine.

4.3 Avoiding or reducing media consumption to handle information overload

The last challenge that participants faced during COVID-19 in obtaining trustworthy information was that the information about the pandemic was too abundant and uncertain to process. For our participants, the sheer amount of data surrounding COVID-19 overwhelmed them, especially at the beginning of the pandemic when they had difficulty in precisely understanding the situation and the disease. The news media were incessantly reporting the rising number of cases and deaths from COVID-19, and the protocols on how to protect individuals against COVID-19 were copious and ever-changing. This exacerbated the confusion among our participants because they were unsure whom to reach out to and what to follow and believe. The situation with their chronic conditions added to this struggle because they had to be extra careful in order to avoid being infected with the disease due to their higher risk.

"We got [infected with] COVID-19 before the CDC changed the guidelines to shorten the period of quarantine and isolation. And that was something that we talked about at our house meeting like, "Do we follow the old guidelines, do we follow the new guidelines?" It's confusing because there's so much information, but there's also so little information that they have sometimes" (P10, female).

As the quote from P10 suggests, some participants revealed that the uncertainty of information on COVID-19 frustrated them. They were confused because the information around COVID-19 kept being revised by the government and the CDC. At the beginning of the pandemic, medical authorities faced difficulties in trying to comprehend the situation, and there were rapid developments in the crisis on a day-to-day basis. Whenever health protocols and instructions from the CDC changed or different guidelines were

suggested, participants felt confused and frustrated, not knowing who and what source to trust.

Participants came up with two main strategies to avoid this information overload. The first strategy was to limit the number of sources they used to obtain trustworthy information. For example, some participants chose to rely on one or two sources of information that they found most trustworthy based on their prior experience. Another strategy was to take a complete pause in their information search and give themselves some rest in order to avoid the stress. Some participants wanted to keep their mental health positive and clear by not letting in negative information and emotions. Thus, they went back to their research on COVID-19 after getting some time away from the information. They dedicated their time and effort because they were keen to actively prevent themselves from becoming infected, considering their high risk.

"I've read that they've talked about numbers (of COVID-19 cases) have been inflated. That's just been word that I've heard on the news... So, I've been limiting just some of my contact to it (COVID-19 information) because I think sometimes I just get overwhelmed with the amount of information. That I just kind of concentrate on a source or two and leave it at that."
(P14, female)

This strategy of limiting the sources of information and their time in doing research on COVID-19 was important because it indicated how participants were active in not only searching for more information but refraining from doing so. They played an active role in juggling their conditions, both physically and mentally, because it was important for them to maintain both. Their chronic conditions were a marathon and COVID-19, while it is dangerous, was also another hurdle that needed to be overcome. Therefore, participants self-monitored their behaviors and mental state while searching for information on COVID-19.

5 LIMITATION AND FUTURE WORK

Thus far, we have described three challenges and strategies that our African American participants with chronic conditions experienced while they were searching for trustworthy information during COVID-19. Participants' active evaluation of the credibility of sources, searching for first-hand experiences from those they could relate to, and avoiding information overload through limiting their time in engaging with COVID-19 information helped them overcome their existing obstacles. We believe that application of these strategies is crucial in developing future technology that engenders trust towards health information by African Americans

Table 2: Summary of the findings

Challenges	Strategies
- Being misled by false or partial information on the internet (e.g., social media)	- Actively evaluating the credibility of a variety of information sources
- COVID-19 information detached from participants' health situation or historical context	- Obtaining first-hand experience by people participants could relate to
- Being overwhelmed by abundant and uncertain information on COVID-19	- Avoiding or reducing media consumption to handle information overload

with chronic conditions. For instance, we can imagine a platform that makes it convenient for the population to evaluate the credibility of diverse sources and at the same time, allows the sharing of experience from people they can relate to.

Our study is not without its limitations. First of all, our participants are limited to African Americans living in a Midwest city in the U.S., and most of the participants were women. Past studies have suggested that African American men have higher levels of medical mistrust than African American women, which may influence how they meet different challenges and come up with other strategies in seeking health information. Secondly, this study was conducted when quarantine was highly encouraged and only a small proportion of the population was allowed to take the COVID-19 vaccines. Now that the situation regarding COVID-19 has developed with different variant viruses and booster shots, how the African American population perceives the current situation and how they use technology may be different.

For future work, we would like to extend this study to actually designing technology that would support African Americans with chronic conditions in obtaining trustworthy information. This technology could range from website-based platforms to chatbots that provide health information. We plan to design a study in which participants use such technology and examine how they were better able to obtain health information than previous methods. They would also be able to evaluate the trustworthiness of the information from the technology and the frequency of usage. Through quantitative and qualitative evaluation, we would like to determine the usefulness and feasibility of the technology for the African American population.

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